Social coding and open software - What can you do to get credit for your code and to allow reuse

In this lesson we will discuss how and why to share code and data, what kind of licenses are used in what situation, and how software can be cited.

We will try to connect software licenses to FAIR principles, give practical recommendations for starting, contributing, and reusing code and help you navigating and deciding on licenses.

Why software licenses matter

- You **find some great code** or data that you want to reuse for your own publication (good for the original author: you will cite them and maybe other people who cite you will cite them).
- You need to **modify the code** a little bit, or you remix the data a bit.
- When it comes time to publish, you realize there is no **license**.

Now we have a problem:

- You manage to **publish the paper without the software/data** but others cannot build on your software and data and you don't get as many citations as you could.
- Or, you **cannot publish it at all** if the journal requires that papers should come with data and software so that they are reproducible.

This lesson is about how to avoid this situation for you and others.

⚙ Prerequisites

No computational prerequisites required for this lesson.

20 min	Social coding
90 min	Software licensing focusing on open source
20 min	Software citation
10 min	Sharing data

Social coding

Objectives

```
## Question 1: Why would I want to share my scripts/code/data?

**Choose many**. Vote by adding an `o` character:

- A: Easier to find and reproduce (scientific reproducibility)
  - votes:

- B: More trustworthy: others can verify correctness and find and report bugs
  - votes:

- C: Enables others to build on top of your code
     (derivative work, provided the license allows it)
  - votes:

- D: Others can submit features/improvements
  - votes:

- E: Others can help fixing bugs
  - votes:

- F: Many tools and apps are free for open source, so no financial cost for this
     (GitHub, GitLab, Appveyor, Read the Docs)
  - votes:

- G: Good for your CV: you can show what you have built
  - votes:

- H: Discourages competitors. If others can't build on your work,
     they will make competing work
  - votes:

- I: When publicly shared, usually we time-stamp or set a version
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     they will make competing work
  - votes:

- I: When publicly shared, usually we time-stamp or set a version,
     so it is easier to refer to a specific version
  - votes:

- J: You can reuse your own code later after change of job or affiliation
  - votes:

- K: It encourages me to code properly from the start
  - votes:


## Question 2: The most concerning thing for me, If I share my software now

**Choose one**. Vote by adding an `o` character:

- A: It will be scooped (stolen) by someone else
  - votes:

- B: It will expose my "ugly code"
  - votes:

- C: Others may find bugs and mistakes. What if the algorithm is wrong?
  - votes:

- D: I will get too many questions, I do not have time for that
  - votes:

- E: Losing control over the direction of the project
  - votes:

- F: Low quality copies will appear
  - votes:

- G: I won't be able to sell this later. Someone else will make money from it
  - votes:

- H: It is too early, I am just prototyping, I will write version to distribute later
  - votes:

- I: Worried about licensing and legal matters, as they are very complicated
  - votes:


## Question 3: Why is software often treated differently from papers?

Free-form answers:
- ...
- ...
- ...
- ...
- ...
- ...


## Question 4: When you find a repository with code/library you would like to reuse,
what are the things you look at to decide whether you use it?

Free-form answers:
- ...
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## Question 3: Why is software often treated differently from papers?

Free-form answers:
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## Question 4: When you find a repository with code/library you would like to reuse,
what are the things you look at to decide whether you use it?

Free-form answers:
- ...
- ...
- ...
- ...
- ...
- ...
```
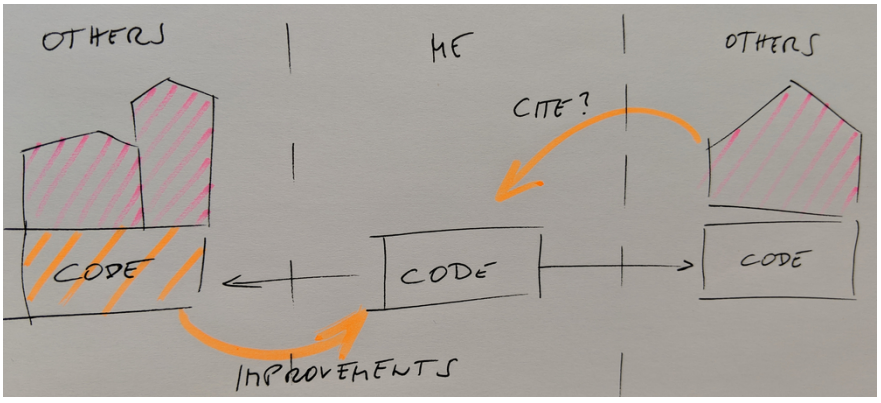

- The goal is maximum visibility and maximum reuse.
- The more interesting science is done referencing my paper, the better for me.
- Nobody actively tries to limit the reach of their papers.



Different ways we can benefit from sharing code.

Sharing code:

- "I did all the ground work and they get to do the interesting science?"
- Sharing code and encouraging *derivative work* may boost your academic impact.
  - But will your work be visible if it is used two levels deep down?

However a study showed that despite these policies, many people still do not share their code 😞. This paper includes samples of charming author responses such as:

Journal policies as motivation for sharing

> "When you approach a PI for the source codes and raw data, you better explain who you are, whom you work for, why you need the data and what you are going to do with it."

From Science editorial policy:

Motivation for open source software

> "We require that **all computer code used for modeling and/or data analysis** that is not [illegible] available be deposited in a **publicly accessible repository** upon publication. In rare exceptional cases where security concerns or competing commercial interests pose a conflict, code-sharing arrangements that still facilitate reproduction of the work should be discussed with your Editor no later than the revision stage."

Instructor note

We revisit answers to Question 1 (above).

- Enable derivative work
  - Do not lock yourself out of own code
- Attract developers who want to be able to show the coding work on their CVs
- Tightly regulated domains require open source
- Open source software (OSS) can lead to more engagement from industry, which may lead to more impact
- If it is open it will likely become standard

From Nature editorial policy:

> "An inherent principle of publication is that others should be able to replicate and build upon the authors' published claims. A condition of publication in a Nature research journal is that authors are required to make **materials, data, code, and associated protocols promptly available** to readers without undue qualifications. Any restrictions on the availability of materials or information must be disclosed to the editors at the time of submission. Any restrictions must also be disclosed in the submitted manuscript."

Sharing software is also scary

Instructor note

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Sharing software is also scary

Instructor note

We revisit answers to Question 2 (above).

- **Fear of being scooped**: A license can avoid it, and you can release when you are ready. Anyway, it is very unlikely that others will understand your code and publish before you without involving you in a collaboration. Sharing is a form of publishing.
- **Exposes possibly "ugly code"**: In practice almost nobody will judge the quality of your code. "Software, once written, is never really finished" (N. Asparouhova).
- **Others may find bugs and mistakes**: Isn't this good? Would you not like to use a code which gives people the chance to locate bugs? If you don't release, people will assume there are bugs anyway.
- **Others may require support and ask too many questions**: This can become a problem: use tools and community and protect your time. You aren't required to support anyone. You can also "archive" a repository to disable most forms of interaction (issues, PRs). Also a note in README on support level helps.
- **Fear of losing control over the direction of the project**: Open source does not mean everybody can change **your version**.
- **"Bad" derivative projects may appear**: It will be clear which is the official version.

Code reusability: What contributes to reusability?

What contributes to you being able to reuse stuff that others make, and others (or you) being able to reuse your stuff? When you find a repository with code you would like to reuse, you may look at the following things to determine its reusability:

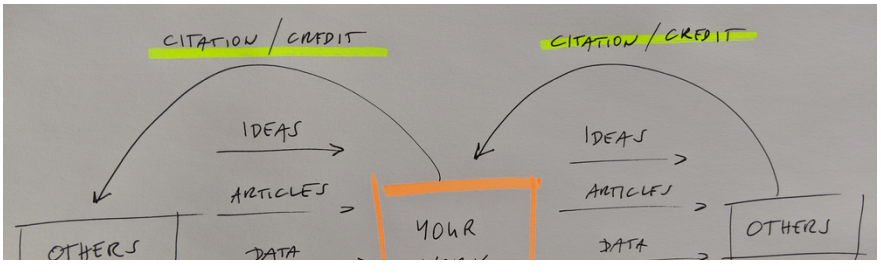
- Have you experienced an implicit expectation of support?
- Supporting all requests can lead to overworking and mental health issues.
- Not supporting requests can also induce guilt.
- Most projects are maintained by 1 or 2 persons.
- Most projects cannot retain contributors for a longer time. Interests change. "Casual contributors are like tourists visiting NYC for a weekend" (Nadia Asparouhova, book below).
- If you maintain all projects that you start forever, at some point it may be difficult to start new projects.
- What are your experiences? Do you agree with the above thoughts?
- Book recommendation: Nadia Asparouhova (formerly Nadia Eghbal): "Working in Public: The Making and Maintenance of Open Source Software (Stripe Press)"

Instructor note

This can be now reconnected to Question 4 (above).

- **Date of last code change**: Is the project abandoned?
- **Release history**: How about stability and backwards-compatibility?
- **Versioning**: Will it be painful to upgrade?
- **Number of open pull requests and issues**: Are they followed-up?
- **Installation instructions**: Will it be difficult to get it running?
- **Example**: Will it be difficult to get started?
- **License**: Are you allowed to do it?
- **Contribution guide**: How to contribute and decision process?

How our work connects to the work of others



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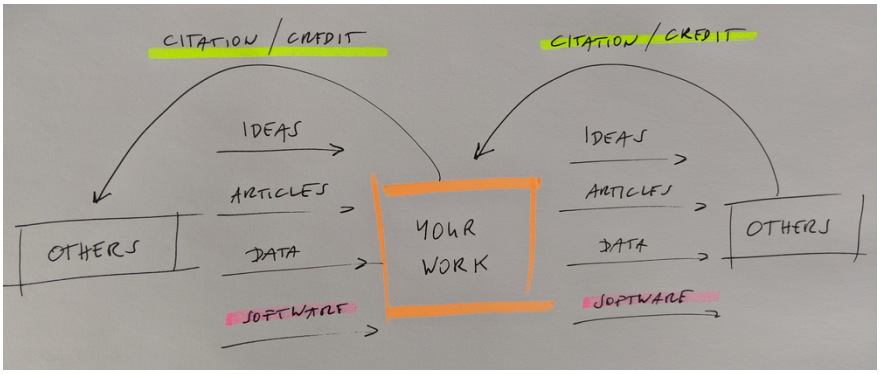

Whether and what we can share depends on how we obtained the components.

- Our work depends on outputs from others. Research of others depends on our outputs.
- Whether you can share your output depends on how you obtained your input.
- A repository that is private today might become public one day.
- Sometimes “OTHERS” are you yourself in the future in a different group/job.
- Our code often starts by changing other code: **derivative work**.
- **Software licenses** matter. And this is what we will discuss in the next episode.

Software licensing focusing on open source

Objectives

- Principles of open source licensing
- Difference between permissive and copyleft licenses
- Regulations for AI-generated and AI-assisted code
- Determine the software license for your project following EU regulation
- Navigate the Joinup Licensing Assistant to select a compliant license
- Understand the licensing distinction between container recipes and container images

Research Software Alliance Policy Directory

This lesson is designed as practical educational material for researchers and research software engineers, **not formal legal advice**

- Regional Focus: Guidance is grounded in EU statutory directives, European institutional frameworks and developers based in Europe with a global focus.
- Institutional Context: Employment contracts, grant agreements, and university policies heavily influence software ownership and licensing choices.
- This lesson covers only the general principles of open-source reuse, copyright scope, and software adaptation.

If you need formal guidance reference below and legal experts at your host institute could be of help.

- EUR Directive 2009/24/EC
- Compendium of U.S. Copyright Office Practices (3rd Ed.) – Chapter 700, Section 721: Computer Programs
- Chinese Regulations on Computer Software Protection (search: “??????????”)
- Joinup Licensing Assistant
- REUSE Initiative

Introduction

In the European Union, software protection is governed by copyright law, which makes a sharp distinction between what is protected and what is not:

- Protected: The specific source code text, expression, binaries, and preparatory design work.
- Not protected: Underlying mathematical algorithms, ideas, programming logic, and interface principles.

Because copyright only protects the expression and not the underlying ideas, developers use licenses to define how that expression can be legally reused.

Scope of this Lesson: What Counts as “Software”?

Across international legal frameworks (such as 17 U.S.C. § 101 and WIPO model provisions), software is commonly defined as a set of instructions to be used directly or indirectly in a computer to bring about a certain result. Under EU statutory law (Directive 2009/24/EC), computer programs—including their preparatory design material—are protected under copyright as literary works.

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Because copyright protection hinges on functional execution combined with creative human expression, this lesson covers the full spectrum of modern research software assets:

- **Source Code**: Original algorithms written from scratch or implemented from scientific papers.
- **Third-Party Integrations**: Embedded permissive or copyleft code snippets and dynamically/statically linked libraries.
- **Container Recipes**: Infrastructure as Code text files (`Dockerfile`, Apptainer `.def`).
- **Container Images**: Bundled binary filesystem snapshots (`.sif` files, OCI registry images).
- **AI-Assisted Code**: Code generated, refactored, or assembled with human creative oversight.
- **AI Prompt Templates**: Complex, engineered system prompts and structured frameworks meeting the threshold of human creative authorship.

Global Context: Software Engineering Across Legal Borders

Software development is an inherently cosmopolitan business. Research software engineers routinely collaborate across continents, fetch dependencies from global registries, and commit code to international repositories.

However, modern developers face a subtle trap: **AI legal bias**. Coding assistants (ChatGPT, Claude, GitHub Copilot) are overwhelmingly trained on US-centric web data and legal texts. Consequently, when asked about software ownership or licensing, AI outputs almost universally default to **US common law concepts** (*"Fair Use"*, *"Work Made for Hire"*, *"Derivative Works"*). Relying blindly on AI advice can create legal blind spots when operating in the EU or collaborating globally.

Deep Dive: Comparative Legal Mechanisms (US vs. EU vs. Asia) ›

Classification of licenses

```
  flowchart LR
subgraph box[ ]
    A["Copyright Law Foundation<br/>(EU Directive 2009/24/EC)"] --> B["Permissive<br/>(MIT, BSD, Apache-2.0)"]
    A --> C["Copyleft / Reciprocal<br/>(EUPL, GPL, LGPL)"]
    A --> D["All Rights Reserved / Proprietary"]

    B --> B1["Run & Modify?<br/><b>Yes!</b>"]
    B --> B2["Sell copies as-is?<br/><b>Yes!</b>"]
    B --> B3["Embed in closed product & sell?<br/><b>Yes!</b>"]
    B --> B4["Must changes stay open?<br/><b>No</b> (Optional)"]

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    subgraph osi["OSI compatible"]
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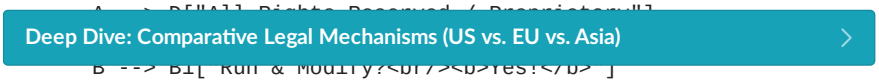


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      classDef permissive fill:#e6ffe6,stroke:#2b8a3e,stroke-width:2px,color:#1b4332;
      classDef copyleft fill:#fff9db,stroke:#f59f00,stroke-width:2px,color:#5c3c00;
      classDef proprietary fill:#ffe3e3,stroke:#e03131,stroke-width:2px,color:#5c0000;
      classDef header fill:#f8f9fa,stroke:#495057,stroke-width:2px,color:#212529;
      classDef mains fill:#fafadc,stroke:#495057,stroke-width:2px,color:#212529;
      classDef osiBox fill:#f8f9fa,stroke:#0275d8,stroke-width:2px,stroke-dasharray: 5 5,color:#0275d8;
      classDef box fill:#ffffff;
      class B1,B2,B3,B4,C1,C2 permissive;
      class C3,C4,D1,D2,D3,D4 proprietary;
      class box box;
      class A,B,C,D mains;
      class osi osiBox;
```

Classification of licenses

Best Practice: In-File Identification using SPDX

Once you select a license, apply it to individual source files and build recipes using **SPDX identifiers** (Software Package Data Exchange). Managed by the Linux Foundation, an SPDX identifier is a standardized, machine-readable short tag (e.g., `MIT`, `Apache-2.0`, `GPL-3.0-only`, `0BSD`) recognized by automated compliance scanners, package managers, and CI/CD build pipelines.

Instead of pasting long legal texts at the top of every file, add a single-line comment at the very first line of your script or recipe:

- 🟢 Can (Rights): What you are allowed to do (e.g., Run, Modify). Matches the "Yes!" bubbles in the diagram.
- ⚪ Must (Obligations): What you are required to do (e.g., Include Copyright for
- In a container recipe

```
# SPDX-License-Identifier: MIT
FROM ubuntu:24.04
```

- 🟡 Support: External verification (e.g., OSI approved). Maps to the blue dashed box.

✍️ Scenario 1: Own algorithm with external dependencies

- In a python script

You wrote an original algorithm from scratch (in Python, C++, Rust, etc.). Your repository

```
# SPDX-License-Identifier: 0BSD
import numpy as np
```

- **Licensing Goal**: You want **maximum adoption** and zero friction for commercial or academic reuse.

How to select a license

Licensing Assistant selection guide:

Lets go through some examples on how to use the European Commission's Joinup Licensing Assistant (JLA) to select licenses. The JLA groups license clauses into four categories that map to the visual diagram above:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved
☑ Modify/merge			
☑ Distribute			

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✔ Solution

Legal Reality: External dependencies remain separate works. Because you have not bundled third-party code inside your repository, you hold full copyright over your original codebase.

- **Outcome**: **Fully Permissible.** You own the code and can choose any open-source license.
- **Selected Category**: **Permissive** (driven by your goal of maximum adoption).
- **JLA Expected Matches**: `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Why**: The filters select licenses granting maximum reuse while requiring only basic copyright attribution (`Incl. Copyright`).
- **User Obligation**: Downstream users must comply with individual external package licenses when fetching, compiling, or running them.
- **Mixing & Redistribution**: Anyone can freely mix, embed, or redistribute your source code. If a user compiles and distributes a combined **binary** that dynamically links to a copyleft shared library (e.g., GPL `.so`), their *distributed compiled binary* must comply with copyleft obligations, but your upstream source repository remains unaffected under your chosen permissive license.

✍️ Scenario 2: Implementing an algorithm from a paper

You read a published scientific paper or technical specification, understand the underlying mathematical algorithm, and write your own original software implementation from scratch.

- **Licensing Goal**: You want **reciprocal protection** anyone can use your implementation, but any downstream modifications distributed by others must remain open source.

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		

✔ Solution

- **Outcome**: **Fully Permissible.** You own 100% of the copyright for your software implementation and can choose any open-source license.
- **Selected Category**: **Copyleft / Reciprocal** (driven by your goal of community protection).
- **JLA Expected Matches**: `EUPL-1.2`, `GPL-3.0`, `AGPL-3.0`
- **Why**: Adding `Copyleft/Share a.` and `Disclose source` under the **Must** column isolates reciprocal terms while leaving all other baseline criteria identical.
- **User Obligation**: Users who redistribute your software or their modified versions must provide source code access under the same copyleft terms.
- **Mixing & Redistribution**: Anyone can use and modify your code. However, if a third party integrates your copyleft implementation into their software and distributes the combined product, their whole application must be released under a compatible open-source copyleft license.

✍️ Scenario 3: Directly embedding third-party Permissive source code

You find a useful helper module online licensed under a **Permissive license** (e.g., MIT or BSD-3-Clause). You copy and paste this code directly into your repository to build upon it.

- **Legal Reality**: Under EU Directive 2009/24/EC Art. 1(2), copyright protects specific source code *expression*, not underlying mathematical algorithms or scientific principles. Writing a fresh implementation creates a brand-new, independent copyright.
- **Licensing Goal**: You want to know if including permissive third-party code limits your overall repository license choices (e.g., if you prefer a Copyleft license like EUPL-1.2 or

- **Outcome**: **Fully Permissible.** You own 100% of the copyright for your software implementation and can choose any open-source license.
- **Selected Category**: **Copyleft / Reciprocal** (driven by your goal of community protection).
- **JLA Expected Matches**: `EUPL-1.2`, `GPL-3.0`, `AGPL-3.0`
- **Why**: Adding `Copyleft/Share a.` and `Disclose source` under the **Must** column isolates reciprocal terms while leaving all other baseline criteria identical.
- **User Obligation**: Users who redistribute your software or their modified versions must provide source code access under the same copyleft terms.
- **Mixing & Redistribution**: Anyone can use and modify your code. However, if a third party integrates your copyleft implementation into their software and distributes the combined product, their whole application must be released under a compatible open-source copyleft license.

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✍️ Scenario 3: Directly embedding third-party Permissive source code

You find a useful helper module online licensed under a **Permissive license** (e.g., MIT or BSD-3-Clause). You copy and paste this code directly into your repository to build upon it.

- **Legal Reality**: Under EU Directive 2009/24/EC Art. 1(2), copyright protects specific source code *expression*, not underlying mathematical algorithms or scientific principles. Writing a fresh implementation creates a brand-new, independent copyright.
- **Licensing Goal**: You want to know if including permissive third-party code limits your overall repository license choices (e.g., if you prefer a Copyleft license like EUPL-1.2 or GPL-3.0).

Licensing Assistant selection guide (example selecting Copyleft):

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✔ Solution

Legal Reality: Permissive licenses (MIT, BSD) grant broad rights to combine, modify, and re-license derivative works under different terms, provided you preserve the original author's copyright notice and license text in the copied files.

- **Outcome**: **Full Flexibility.** Unlike inbound Copyleft (Scenario 4), embedding Permissive code does not force a specific license on your project. You can license your combined repository as Permissive *or* Copyleft.
- **Selected Category**: **Copyleft** (or Permissive, depending on your intent).
- **JLA Expected Matches**: `EUPL-1.2`, `GPL-3.0` (or `MIT`, `Apache-2.0` if Permissive goal).
- **Why**: Permissive inbound code is compatible with almost all OSI-approved software licenses.
- **User Obligation**: You must retain the original copyright notice and MIT/BSD license text within the specific files or NOTICE file where the copied code resides.
- **Mixing & Redistribution**: Downstream users follow your repository's overall license terms, but the original permissive author's attribution notice must remain intact inside the codebase.

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✔ Solution

Legal Reality: Unlike referencing external dependencies or writing code from scratch, pasting third-party source code directly into your repository creates a single combined (derivative) work. You do not hold exclusive copyright over the entire codebase.

✍️ Scenario 4: Directly embedding third-party Copyleft source code

You find a useful utility function or module online licensed under a **Copyleft / Reciprocal license** (e.g., GPL-3.0 or EUPL-1.2). You copy and paste this source code directly into your repository files and extend it to fit your project.

- **Licensing Goal**: Fulfill legal requirements by incorporating inbound copyleft code into your codebase.
- **Selected Category**: **Copyleft / Reciprocal** (mandated by the inbound license's copyleft clause).
- **JLA Expected Matches**: `EUPL-1.2` `GPL-3.0`
- **Why**: Inbound copyleft terms mandate that any derivative work distributed as a whole must inherit reciprocal sharing obligations (`Copyleft/Share a.` and `Disclose source`).

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved

- **User Obligation**: Anyone distributing your project must provide access to the full source code (including your modifications) under the matching copyleft terms.
- **Mixing & Redistribution**: Downstream users receive full copyleft freedoms. You cannot re-license your combined repository under a permissive license later unless

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

- **Why**: Permissive inbound code is compatible with almost all OSI-approved software licenses.
- **User Obligation**: You must retain the original copyright notice and MIT/BSD license text within the specific files or NOTICE file where the copied code resides.

✔ Solution

- **Mixing & Redistribution**: Downstream users follow your repository's overall license terms, but the original permissive author's attribution notice must remain intact inside the codebase.

Legal Reality: Unlike referencing external dependencies or writing code from scratch, pasting third-party source code directly into your repository creates a single combined (derivative) work. You do not hold exclusive copyright over the entire codebase.

✍ Scenario 4: Directly embedding third-party Copyleft source code

You find a useful utility function or module online licensed under a **Copyleft / Reciprocal license** (e.g., GPL-3.0 or EUPL-1.2). You copy and paste this source code directly into your repository files and extend it to fit your project.

- **Licensing Goal**: Publish legally by incorporating inbound copyleft code into your codebase.
- **JLA Expected Matches**: `EUPL-1.2` and `GPL-3.0`
- **Why**: Inbound copyleft terms mandate that any derivative work distributed as a whole must inherit reciprocal sharing obligations (`Copyleft/Share a.` and `Disclose source`).

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✔ Solution

- **Outcome**: **Restricted Choice (Mandatory Copyleft).** You cannot choose a permissive license (e.g., MIT) or keep the repository proprietary. You must choose a copyleft license compatible with the inbound code.
- **Selected Category**: **Copyleft / Reciprocal** (mandated by the inbound license's copyleft clause).
- **JLA Expected Matches**: `EUPL-1.2` , `GPL-3.0` , `AGPL-3.0`
- **Why**: The inbound copyleft terms mandate that any derivative work distributed as a whole must inherit reciprocal sharing obligations (`Copyleft/Share a.` and `Disclose source`).
- **User Obligation**: Anyone distributing your project must provide access to the full source code (including your modifications) under the matching copyleft terms.
- **Mixing & Redistribution**: Downstream users receive full copyleft freedoms. You cannot re-license your combined repository under a permissive license later unless you completely strip out or rewrite the third-party copyleft code from scratch.

✍ Scenario 5: Linking against a Strong Copyleft library (e.g., GSL or FFTW)

You write your own original code from scratch, but your program includes or links against a third-party scientific library licensed under a **Strong Copyleft license** (such as GPL-3.0).

- **Licensing Goal**: You want to publish your repository and need to select a license that complies with the inbound linking requirements of the GPL library.

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✔ Solution

Legal Reality: Linking your code (statically or dynamically) with a Strong Copyleft library like GPL creates a combined software work upon compilation and distribution. The strong copyleft obligation extends across the linking boundary.

- **Outcome**: **Mandatory Copyleft.** To distribute the compiled application or repository, your code must be licensed under a GPL-compatible copyleft license. You cannot license the overall project under a Permissive license (like MIT).
- **Selected Category**: **Copyleft / Reciprocal** (required by the linked GPL library).
- **JLA Expected Matches**: `GPL-3.0` , `AGPL-3.0` , `EUPL-1.2`
- **Why**: The linked library's reciprocal license mandates that any distributed program depending on it must also provide source code access under compatible copyleft terms (`Copyleft/Share a.` and `Disclose source`).
- **User Obligation**: Users who distribute binaries or modified packages of your project must provide the full source code under the GPL-compatible copyleft license.
- **Mixing & Redistribution**: Anyone using or building upon your work must maintain the GPL-compatible copyleft license. If you want to avoid copyleft restrictions for your codebase, you must replace the GPL library dependency with a permissively licensed alternative (e.g., an MIT or BSD library).

✍ Scenario 6: Using AI coding assistants (e.g., GitHub Copilot, ChatGPT)

You write software using AI coding assistants (e.g., GitHub Copilot, ChatGPT) to generate functions, boilerplate, or refactor algorithms. Your repository consists of a mix of human-authored code and AI-generated outputs.

- **Licensing Goal**: You want **maximum adoption** (or any open-source model) and need to know if using AI tools restricts your choice of open-source license.

Licensing Assistant selection guide (example using Permissive selection):

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved
☑ Modify/merge			
☑ Distribute			

✔ Solution

Legal Reality: Under EU copyright law and international consensus, **pure AI-generated outputs lacking human authorship are generally ineligible for copyright protection**. However, when you assemble, refine, and integrate AI code into an overarching software project through creative human effort, you hold copyright over the resulting human-authored work (provided the AI tool did not reproduce substantial copyrighted

library like GPL creates a combined software work upon compilation and distribution. The strong copyleft obligation extends across the linking boundary.

✍️ Scenario 6: Generating or assisting code using AI tools

You write software using AI coding assistants (e.g., GitHub Copilot, ChatGPT) to generate functions, boilerplate, or refactor algorithms. Your repository consists of a mix of human-authored code and AI-generated outputs.

- **Licensing Goal**: You want **maximum adoption** (or any open-source model) and need to know if using AI tools restricts your choice of open-source license.
- **Selected Category**: **Copyleft / Reciprocal** (required by the linked GPL library).
- **JLA Expected Matches**: `GPL-3.0`, `AGPL-3.0`, `EUPL-1.2`
- **Why**: The linked library's reciprocal license mandates that any distributed program depending on it must also provide source code access under compatible copyleft terms (`Copyleft/Share a.` and `Disclose source`).
- **User Obligation**: Users who distribute binaries or modified packages of your project must provide the full source code under the GPL-compatible copyleft license.
- **Mixing & Redistribution**: Anyone using or building upon your work must maintain the GPL-compatible copyleft license. If you want to avoid copyleft restrictions for your codebase, you must replace the GPL library dependency with a permissively licensed alternative (e.g., an MIT or BSD library).

✔ Solution

Legal Reality: Under EU copyright law and international consensus, **pure AI-generated outputs lacking human authorship are generally ineligible for copyright protection**. However, when you assemble, refine, and integrate AI code into an overarching software project through creative human effort, you hold copyright over the resulting human-authored work (provided the AI tool did not reproduce substantial copyrighted third-party snippets verbatim).

- **How Much AI Assistance Is Allowed**: There is no fixed percentage threshold. If a legal dispute arises, courts evaluate **Human Authorship and Creative Control**. Using AI as a boilerplate code generator, advanced autocomplete, or research assistant where you actively guide, review, modify, and structure the code preserves your copyright ownership. Conversely, simply pressing a button to generate an entire project without human creative intervention yields uncopyrightable output.
- **How to Check for Copyrighted Material**: Combine manual codebase searches (e.g., GitHub Code Search), built-in AI tool filters (such as *Block suggestions matching public code* in GitHub Copilot), and automated open-source license scanners (like FOSSology or Snyk).
- **Outcome**: **Fully Permissible.** The use of AI tools does not force a specific open-source license onto your repository. You retain the choice between Permissive or Copyleft based on your strategic goals.
- **Selected Category**: **Permissive** (or Copyleft, determined by author intent rather than the AI tool).
- **JLA Expected Matches**: `MIT`, `Apache-2.0`, `BSD-3-Clause` (or `EUPL-1.2`, `GPL-3.0` if your intent is Copyleft).
- **User Obligation**: Standard obligations apply based on the license you choose to attach to your human-authored codebase.
- **Mixing & Redistribution**: Anyone can use, modify, or redistribute your repository under your chosen license. Downstream users are bound by your overall repository license terms, while the standalone, raw unedited AI snippets themselves remain ineligible for copyright protection.

✔ Solution

✍️ Scenario 7: Distributing a Container Build Recipe (Dockerfile or Apptainer .def)

You write or generate a container build recipe (`Dockerfile` or Apptainer `.def` file) to make your research reproducible. The recipe contains text commands that pull a base image (`Dockerfile`), install system packages (`apt-get`), clone code from GitHub, and download data.

- **Licensing Goal**: You want **maximum adoption** for your build recipe and zero restrictions on who can use or modify your setup instructions.
- **Selected Category**: **Permissive** (driven by your goal of maximum adoption).
- **JLA Expected Matches**: `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Why**: The recipe file itself is source code. Referencing third-party packages in `RUN` or `FROM` steps is legally equivalent to writing an `import` statement or listing dependencies in `requirements.txt`.
- **User Obligation**: Users who download your recipe file must preserve your copyright notice.
- **Mixing & Redistribution**: Anyone can freely share or modify your `Dockerfile` or `.def` file under your chosen permissive license, regardless of whether the tools installed by the recipe are Permissive, Copyleft, or Proprietary.

Licensing Assistant ↗ selection guide (example using Permissive selection):

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved
☑ Modify/merge			
☑ Distribute			

✔ Solution

Legal Reality: A container recipe is a text file containing build instructions (Infrastructure as Code). Referencing external base images, packages, or repositories in build commands does not transfer third-party copyright onto your text file.

- **Outcome**: **Fully Permissible.** You own the copyright to the build instructions you write and can choose any license for your recipe file.
- **If Generated by AI**: Using AI tools (Copilot, ChatGPT) to generate or refactor a `Dockerfile` follows standard AI code rules. As long as you review, adapt, and configure the recipe for your project, you hold the copyright and retain total freedom over its license.

✍️ Scenario 7: Distributing a Container Build Recipe (Dockerfile or Apptainer .def)

You write or generate a container build recipe (`Dockerfile` or Apptainer `.def` file) to make your research reproducible. The recipe contains text commands that pull a base image, install system packages (`apt-get`), clone code from GitHub, and download data.

- **Licensing Goal**: You want **maximum adoption** for your build recipe and zero restrictions on who can use or modify your setup instructions.
- **Selected Category: Permissive** (driven by your goal of maximum adoption).
- **JLA Expected Matches**: `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Why**: The recipe file itself is source code. Referencing third-party packages in `RUN` or `FROM` steps is legally equivalent to writing an `import` statement or listing dependencies in `requirements.txt`.
- **User Obligation**: Users who download your recipe file must preserve your copyright notice.
- **Mixing & Redistribution**: Anyone can freely share or modify your `Dockerfile` or `.def` file under your chosen permissive license, regardless of whether the tools installed by the recipe are Permissive, Copyleft, or Proprietary.

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved
☑ Modify/merge			
☑ Distribute			

✍️ Scenario 8: Distributing a Built Container Image (Docker Hub or Apptainer .sif)

You build a complete container runtime image (as an Apptainer `.sif` file or an image pushed to Docker Hub/GitHub Container Registry). The compiled image contains a base Linux OS, installed system libraries, runtime dependencies, and your application code.

- **Licensing Goal**: Fulfill legal obligations imposed by distributing a bundled, compiled binary filesystem image containing third-party works.

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Copyleft/Share a.	☑ For software	☑ OSI approved
☑ Modify/merge	☑ Disclose source		
☑ Distribute			

✔ Solution

Legal Reality: Unlike a text recipe file, a compiled container image (`.sif` or registry image) is a **bundle of third-party software works**. You do not hold exclusive copyright over the entire image filesystem.

- **Outcome**: **Mandatory Compliance (Restricted Choice).** You cannot assign a single permissive license to the distributed image. Distribution is governed by the overlapping terms of all installed base layers, packages, and linked binaries inside.
- **Selected Category**: **Copyleft / Reciprocal** (if your application links against or incorporates Copyleft components inside the container).
- **JLA Expected Matches**: `EUPL-1.2`, `GPL-3.0`
- **Why (Linking vs. Aggregation)**: If your application links against or embeds a Strong Copyleft library (`GPL-3.0`) installed in the container, distributing that image triggers copyleft disclosure obligations for your compiled application. However, if GPL components in the image are merely independent system utilities or standalone tools, GPL's "mere aggregation" provisions apply—the GPL license governs those specific tools, but does not extend to your independent application binaries.
- **User Obligation**: Anyone distributing the built image file must ensure compliance with all third-party licenses inside the container, including providing source access for any GPL components or linked works contained in the layers.

✍️ Scenario 9: Including AI prompt templates in LLM applications

You develop a research software pipeline that uses Large Language Models (LLMs) for automated data extraction. Your repository contains Python scripts alongside a `prompts/` directory containing both short functional prompts (e.g., *"Extract keywords from this paper"*) and complex, 500-word structured prompt templates (e.g., system prompts, JSON schemas, and chain-of-thought frameworks).

- **Licensing Goal**: You want **maximum adoption** for your application and want to ensure your prompt templates are legally covered under the same open-source license as your Python code.

Licensing Assistant selection guide:

🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved

over the entire image filesystem.

- **Mixing & Redistribution**: Downstream users who pull your image must abide by individual component licenses. To keep your application source code unencumbered, ensure your app links only against permissively licensed libraries inside the container layers.
- **Outcome**: **Mandatory Compliance (Restricted Choice).** You cannot assign a single permissive license to the distributed image. Distribution is governed by the overlapping terms of all installed base layers, packages, and linked binaries inside.
- **Selected Category**: **Copyleft / Reciprocal** (if your application links against or incorporates copyleft components inside the container).
- **JLA Expected Matches**: `EUPL-1.2`, `GPL-3.0`
- **Why (Linking vs. Aggregation)**: If your application links against or embeds a Strong Copyleft library (`GPL-3.0`) installed in the container, distributing that image triggers copyleft disclosure obligations for your compiled application. However, if GPL components in the image are merely independent system utilities or standalone tools, GPL's "mere aggregation" provisions apply—the GPL license governs those specific tools, but does not extend to your independent application binaries.
- **User Obligation**: Anyone distributing the built image file must ensure compliance with all third-party licenses inside the container, including providing source access for any GPL components or linked works contained in the layers.

✍ Scenario 9: Including AI prompt templates in LLM applications

You develop a research software pipeline that uses Large Language Models (LLMs) for automated data extraction. Your repository contains Python scripts alongside a `prompts/` directory containing both short functional prompts (e.g., *"Extract keywords from this paper"*) and complex, 500-word structured prompt templates (e.g., system prompts, JSON schemas, and chain-of-thought frameworks).

- **Licensing Goal**: You want **maximum adoption** for your application and want to ensure your prompt templates are legally covered under the same open-source license as your Python code.

Licensing Assistant selection guide:

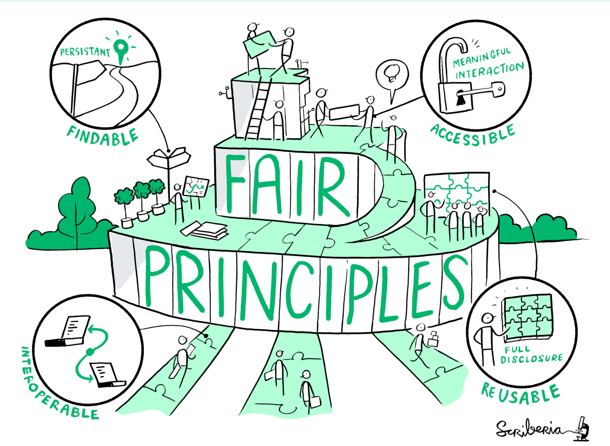
🟢 Can	⚪ Must	🔵 Compatible	🟡 Support
☑ Commercial use	☑ Incl. Copyright	☑ For software	☑ OSI approved
☑ Modify/merge			
☑ Distribute			

✔ Solution

Legal Reality: Copyrightability depends on creative complexity. Simple, functional prompts lack the minimal threshold of creative human expression and carry no copyright. However, complex, highly structured prompt templates are legally classified as literary text assets and are fully protected by copyright.

- **Outcome**: **Fully Coverable.** Engineered prompt templates checked into your repository are treated like source code assets. Applying your overall repository license automatically covers these prompt files.
- **Simple vs. Engineered Prompts**: Short commands (e.g., *"Fix this Dockerfile"*) are uncopyrightable instructions. Complex system prompts, XML-formatted templates, or multi-step reasoning frameworks meet the threshold of creative human authorship.
- **Selected Category**: **Permissive** (driven by your goal of maximum adoption).
- **JLA Expected Matches**: `MIT`, `Apache-2.0`, `BSD-3-Clause`
- **Why**: Permissive open-source licenses cover software text assets, allowing downstream developers to incorporate, modify, and execute your prompt templates in their own AI pipelines.
- **User Obligation**: Standard attribution obligations apply. Downstream users who copy your prompt files must preserve your copyright notice and file headers (e.g., ... pt your prompt ... plications, provided ... plate files.

Is putting software on GitHub/GitLab/... publishing?



FAIR principles. (c) Scriberia for The Turing Way, CC-BY.

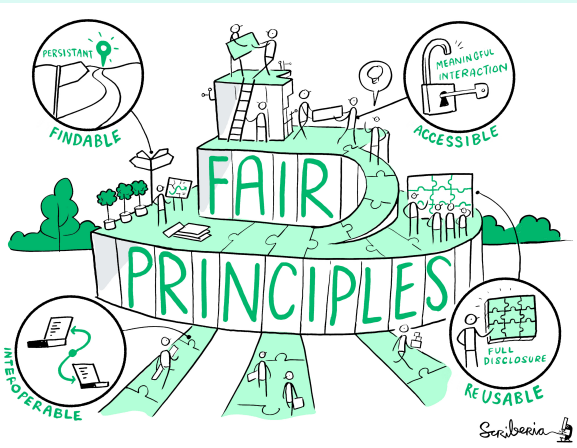
Is it enough to make the code public for the code to remain **findable and accessible**?

- No. Because nothing prevents me from deleting my GitHub repository or rewriting the Git history and we have no guarantee that GitHub will still be around in 10 years.

Is putting software on GitHub/GitLab/... publishing?

User Obligation: Standard Copyright Obligations Apply. Downstream users who copy your prompt files must preserve your copyright notice and file headers (e.g.,

pt your prompt
pplications, provided
plate files.



FAIR principles. (c) Scriberia for The Turing Way, CC-BY.

Is it enough to make the code public for the code to remain **findable and accessible**?

- No. Because nothing prevents me from deleting my GitHub repository or rewriting the Git history and we have no guarantee that GitHub will still be around in 10 years.
- **Make your code citable and persistent**: Get a persistent identifier (PID) such as DOI in addition to sharing the code publicly, by using services like Zenodo or similar services.

How to make your software citable

💬 Discussion (Citation-1): Explain how you currently cite software

- Do you cite software that you use? How?
- If I wanted to cite your code/scripts, what would I need to do?

Checklist for making a release of your software citable:

- ✓ Assigned an appropriate license
- ✓ Described the software using an appropriate metadata format
- ✓ Clear version number
- ✓ Authors credited
- ✓ Procured a persistent identifier
- ✓ Added a recommended citation to the software documentation

This checklist is adapted from: N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Developers (Version 0.9.0). Zenodo. 2019b. (DOI)

Our practical recommendations:

- Web form to create, edit, and validate CITATION.cff files
- Video: "How to create a CITATION.cff using cffinit"
- Add a file called CITATION.cff (example).
- Get a digital object identifier (DOI) for your code on Zenodo (example).
- Step-by-step guide for how to make your GitHub project citable using Zenodo
- Make it as easy as possible: clearly say what you want cited

Papers with focus on scientific software

Where can I publish papers which are primarily focused on my scientific software? Great list/summary is provided in this blog post: "In which journals should I publish my software?" (Neil P. Chue Hong)

How to cite software

This is an example of a simple `CITATION.cff` file:

```yaml
cff-version: 1.2.0
message: "If you use this software, please cite it as below."
authors:
  - family-names: Doe
    given-names: Jane
    orcid: https://orcid.org/1234-5678-9101-1121
title: "My Research Software"
version: 2.0.4
doi: 10.5281/zenodo.1234
date-released: 2021-08-11
```

More about `CITATION.cff` files:

- GitHub now supports CITATION.cff files
- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Authors (Version 0.9.0). Zenodo. 2019a. (DOI)
- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Developers (Version 0.9.0). Zenodo. 2019b. (DOI)

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This is an example of a simple `CITATION.cff` file:

```
cff-version: 1.2.0
message: "If you use this software, please cite it as below."
authors:
  - family-names: Doe
    given-names: Jane
    orcid: https://orcid.org/1234-5678-9101-1121
title: "My Research Software"
version: 2.0.4
doi: 10.5281/zenodo.1234
date-released: 2021-08-11
```

More about `CITATION.cff` files:

- GitHub now supports CITATION.cff files

Recommended format for software citation is to ensure the following information is provided as part of the reference (from Katz, Chue Hong, Clark, 2021 which also contains software citation examples):

- Creator
- Title
- Publication venue
- Date
- Identifier
- Version
- Type

FAIR for Research Software (FAIR4RS)

The FAIR4RS Principles adapt the original FAIR data principles to improve the **Findability, Accessibility, Interoperability, and Reusability** of research software. They provide guidelines to make research software more open, discoverable, citable, and reusable, which supports reproducibility and open science. The summary of the FAIR4RS below is a compressed version of the principles listed in Barker, Michelle, et al. "Introducing the FAIR Principles for research software." Scientific Data 9.1 (2022): 622.

- **Findable**: Software should have a globally unique and persistent identifier (e.g. versioning of releases, DOI from Zenodo when depositing a software release, software metadata in citation file).
- **Accessible**: Software and metadata should be retrievable via standardized protocols (e.g. downloading from GitHub or Zenodo).
- **Interoperable**: Software should use open formats and standards to work with other tools (e.g. input/output files in CSV, dependencies listed in `requirements.txt` or `environment.yml`, configuration files in yaml).
- **Reusable**: Software should have clear licensing, documentation, and provenance (e.g. a standard license and a `README` with usage instructions, authors listed with ORCIDs).

There are great resources to self-evaluate the *FAIRness* of your research software:

- FAIR Software Checklist which provides a questionnaire and even a badge of how FAIR the software is
- FAIR Software NL highlights with nice visuals the five most important elemnts of FAIR4RS (1. Public accessible repository with version control; 2. License; 3. Software registry; 4. Software citation file; 5. Software quality checklist)

Sharing data

Services for sharing and collaborating on research data

To find a research data repository for your data, you can search on the Registry of Research Data Repositories (re3data) platform and filter by country, content type, discipline, etc.

International:

- Zenodo: A general-purpose open access repository created by OpenAIRE and CERN. Integration with GitHub, allows researchers to upload files up to 50 GB.
- Figshare: Online digital repository where researchers can preserve and share their research outputs (figures, datasets, images and videos). Users can make all of their research outputs available in a citable, shareable and discoverable manner.
- EUDAT: European platform for researchers and practitioners from any research discipline to preserve, find, access, and process data in a trusted environment.
- Dryad: A general-purpose home for a wide diversity of datatypes, governed by a nonprofit membership organization. A curated resource that makes the data underlying scientific publications discoverable, freely reusable, and citable.
- The Open Science Framework: Gives free accounts for collaboration around files and other research artifacts. Each account can have up to 5 GB of files without any problem,

Papers with focus on scientific software

Where can I publish papers which are primarily focused on my scientific software? Great list/summary is provided in this blog post: "In which journals should I publish my software?" (Neil P. Chue Hong)

How to cite software

More about `CITATION.cff` files:

- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Authors (Version 0.9.0), Zenodo. 2019a. (DOI)
- N. P. Chue Hong, A. Allen, A. Gonzalez-Beltran, et al., Software Citation Checklist for Developers (Version 0.9.0), Zenodo, 2019b. (DOI)

Sharing data

Services for sharing and collaborating on research data

To find a research repository for your data, you can search the Registry of Research Data Repositories (re3data)[] platform and filter by country, content type, discipline, etc.

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- The Open Science Framework[]: Gives free accounts for collaboration around files and other research artifacts. Each account can have up to 5 GB of files without any problem, and it remains private until you make it public.

Sweden:

- Researchdata.se - research data from a wide range of disciplines[]
- ICOS for climate data[]
- Bolin center climate / geodata[]
- NBIS for life science, sequence, omics data[]

Norway:

- NIRD archive is widely used by some communities[]
- NSD - Norwegian Center for Research Data, for any kind of data[]
- Dataverse.no - Dataverse network, based at University of Tromsø but open for other institutions[]
- ELIXIR Norway for life science, sequence, omics data[]

Denmark:

- Datavejviser[] – metadata catalogue aggregating from research organisations and National Archives
- GEUS Data Hub[] – data from the Geological Survey of Denmark and Greenland
- DTU Data[] – repository of the Technical University of Denmark
- DCE Dataverse for University of Copenhagen[]

Finland:

- IDA, for general data[]
- FSD for social data[]
- more information on Finnish resources for open data[]

Portugal:

- DMPortal - Dataverse network for any research data[]

Licensing of datasets and databases

- The EU has a database directive[] which restricts data mining on databases.
- Has a somewhat similar effect to copyright, because copyright would not apply to data mining.
- A good license also gives rights to data mine. So not a major concern.

When you can use datasets:

- The license allows
- Your country has exceptions for research
- The data doesn't come from the EU

License text, slides, images, and supporting information under a Creative Commons license[], and get a DOI using Zenodo[] or Figshare[] or OSF[] other services.

Resources for data management

Licensing and machine learning/ AI

- DataLad[]
- Data Stewardship Wizard[]

This section is maybe more relevant to **developers of AI models / AI systems** rather than **users of AI models / AI systems**.

Licensing of datasets and databases

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- DTU Data – repository of the Technical University of Denmark
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Is it data? Is it software? It depends. We need to consider the AI system as a whole, the training data, the production data, the AI output, and how it is put on service. **AI models** are like the engine of the car: they cannot do anything without the rest of the car infrastructure. **AI systems** are the whole car with the AI model and all the software and hardware to actually use it.

Depending on what you are going to share, there might be things to consider beyond the license.

For example **large language models** are often shared with open source software licenses, on **HuggingFace** which is like a GitHub/GitLab for AI models (see for example the OLMO model). Many so-called *open-source* models are actually just *open-weights* models: only the trained neural network weights are shared, while the training data, training code, and full documentation are often kept private. This lack of transparency raises concerns about reproducibility and accountability and this phenomenon is sometimes called **"open washing"** (ref). Models are also shared with a **model card** which is a documentation tool for transparency that provide a comprehensive snapshot of a model's characteristics and ethical considerations (see Ch.8 Glerean 2025).

What about ethics? What about liability?

As AI models (e.g. the deep network weights) and AI systems (the model with all the software and infrastructure to query it) are becoming more available, there can be legal (and ethical!) requirements on the developer of the AI model/system by the EU AI Act. In general researchers do not need to worry, but ethically one should consider that if the research-purpose AI model/system could be used for something harmful, ethically (if not legally) one should consider if such model/system should be implemented at all.

What about the training data inside the model? Large models can memorize and unintentionally reveal parts of their training data. This raises concerns about copyright, trade secrets, and personal data. News publishers and artists are suing AI companies for unauthorized use of their content in training. It is still unclear how traditional data licenses can apply to data that has been transformed into model weights.

Further reading

- The Turing Way
- Illustrations from the Turing Way book dashes
- Reproduciblity syllabus
- The reproducible research data analysis platform
- Good talks on open reproducible research can be found here
- "FAIR enough"
- "A FAIker future"
- "Top 10 FAIR Data & Software Things" are brief guides that can be used by the research community to understand how they can make their research (data and software) more FAIR.

More resources

- Five recommendations for fair software
- RAIL initiative: "Responsible AI licenses"
- Publishing research software – A MIT libraries webpage on why to publish software, where to publish software, and how to make software citable.
- The Turing Way: Machine Learning Model Licenses
- "Expert Q&A on Artificial Intelligence (AI) Licensing"
- Software Quality Checklist
- MolSSI Best Practice Guides
- Five recommendations for fair software
- Awesome Research Software Registries

Who is the course for?

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- Awesome Research Software Registries

Who is the course for?

- Someone writing their own relatively small material
- Someone who wants to incorporate other code/libraries into their own projects
- Group leader who wants to decide how to balance openness and long-term strategy

List of exercises

Full list

This is a list of all exercises and solutions in this lesson, mainly as a reference for helpers and instructors. This list is automatically generated from all of the other pages in the lesson. Any single teaching event will probably cover only a subset of these, depending on their interests.

Instructor guide

Basics

This is basically presented as-is - not much technical presentation is needed.

There are the following main acts:

- First section is about social coding and sharing code and motivation for sharing code with a discussion that licensing alone isn't enough. If you want people to use your work, you need to do more to make your work easily reusable.
- Emphasis of the concept of derivative work and that everything is based on other things.
- And if your goal is citations, you want as many people using your work as possible, because then they can cite you.
- Overview of license types. There are many online resources, and the biggest question is permissive vs. weak copyleft vs. strong copyleft. The cake analogy for licenses is only linked and can be skipped and does not have to be discussed in detail.
- Then a brief section on code citations. It's good if you manage to start a discussion around this topic.
- The "Sharing data" part is meant to bring awareness about FAIRness and Open Science and to show how DOIs can be obtained for Git repositories.

This talk used to be called "Software licensing" you could mention that to make it more interesting, we have adopted the current broader name, and our emphasis is not just licensing, but why licensing is needed.

Software licensing

Objectives

- Knowing about what derivative work is and whether we can share it.
- Get familiar with terminology around licensing.
- Practical advice for software licensing.

Copyright

Wi

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Copyright

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- **Trademark**: Protects a name/brand from impersonation.
- **Patent**: Protects a novel, non-obvious, technical invention.
- **Copyright**: Protects **creative expression**: software, writing, graphics, photos, certain datasets, this presentation. Practically "forever" (lifetime of author + 70 years).

Copyright controls whether and how we can distribute the original work or the **derivative work**.

Derivative work: Sampling/remixing

[Midjourney, CC-BY-NC 4.0]

- Changing and distributing software is similar to changing and distributing music
- You can do almost anything if you don't distribute it

Often we don't have the choice:

[Midjourney, CC-BY-NC 4.0]

- We are expected to publish software
- Sharing can be seen as insurance against being locked out

Exercise: Derivative work

💬 Licensing-1: What constitutes derivative work?

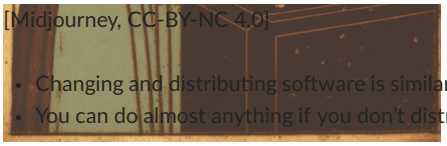
This question 5 below can be used as a starting point and copied to the collaborative document platform input or an online poll:

```
## Question 5: Which of these are derivative works?

**Choose many**. Vote by adding an `o` character:

- A. Download some code from a website and add on to it
  - votes:

- B. Download some code and use one of the functions in your code
```



[Midjourney, CC-BY-NC 4.0]

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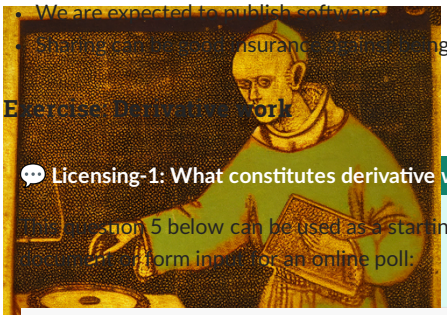

[Midjourney, CC-BY-NC 4.0]

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- Sharing can be seen as insurance against being locked out

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**Choose many**. Vote by adding an `o` character:

- A. Download some code from a website and add on to it
  - votes:

- B. Download some code and use one of the functions in your code
  - votes:

- C. Changing code you got from somewhere
  - votes:

- D. Extending code you got from somewhere
  - votes:

- E. Completely rewriting code you got from somewhere
  - votes:

- F. Rewriting code to a different programming language
  - votes:

- G. Linking to libraries (static or dynamic), plug-ins, and drivers
  - votes:

- H. Clean room design (somebody explains you the code but you have never seen it)
  - votes:

- I. You read a paper, understand algorithm, write own code
  - votes:
```

✔ Solution

- Derivative work: A-F
- Not derivative work: G-I
- E and F: This depends on how you do it, see clean room design.

copyright anymore). Copyright infringment instead is illegal and it can result in criminal charges (e.g. fines). Copyright however protects the particular expression of an idea or fact (for example, the specific source code of a program, but not the underlying algorithm itself).

This insert can be skipped and left as reading exercise

There is no pre-defined "number of lines of code", "seconds of a song", or "pixels of an image" that can clearly set the basis for plagiarism or IPR infringement. However in the context of research, it can be possible to use *Quotation Exception* (in EU, ref) and *Fair use* (in USA, ref). Fair use has become controversial recently as it is used as legal basis for training large language models based on scraped internet data (See for example Henderson, P., Li, X., Jurafsky, D., Hashimoto, T., Lemley, M. A., & Liang, P. (2023). Foundation models and fair use. Journal of Machine Learning Research, 24(400), 1-79.)

Derivative work and containers

Containers are a bit more tricky when it comes to licenses.

- Distribution of container recipes: it's like distributing source code
- Distribution of container images: it can be considered like distributing a binary compiled software

The latter case is a bit more nuanced and the interested reader should read more about "Mere Aggregation" at GPL FAQ. Briefly, if the container image just bundles separate

In academic context it is important to consider also *plagiarism* and how it relates to copyright and more broadly Intellectual Property Rights (a clear explanation at this page). Plagiarism is the practice of taking somebody else's ideas or work and claim them as your own: it is the **unacknowledged** use of another person's work. Intellectual Property Rights (IPRs) infringement instead is the **unauthorised** use of another's work.

IPRs can be classified in two main groups (WTO): i) Copyright and rights related to copyright (computer programs are here) and ii) Industrial properties like trademarks, and inventions (which may include specific technical implementations of systems or code) protected by patents.

In research ethics, plagiarism is one of the three definition of research misconduct (along with *fabrication* and *falsification*, see ALLEA, European Code of Conduct for Research Integrity). Plagiarism is not illegal per se, but it can lead to serious consequences like the retraction of published work. One can engage in plagiarism, without necessarily breaking any IPR law (e.g. write a new book by reusing the plot of an old book that is not under

copyright anymore). Copyright infringment instead is illegal and it can result in criminal charges (e.g. fines). Copyright however protects the particular expression of an idea or fact (for example, the specific source code of a program, but not the underlying algorithm itself).

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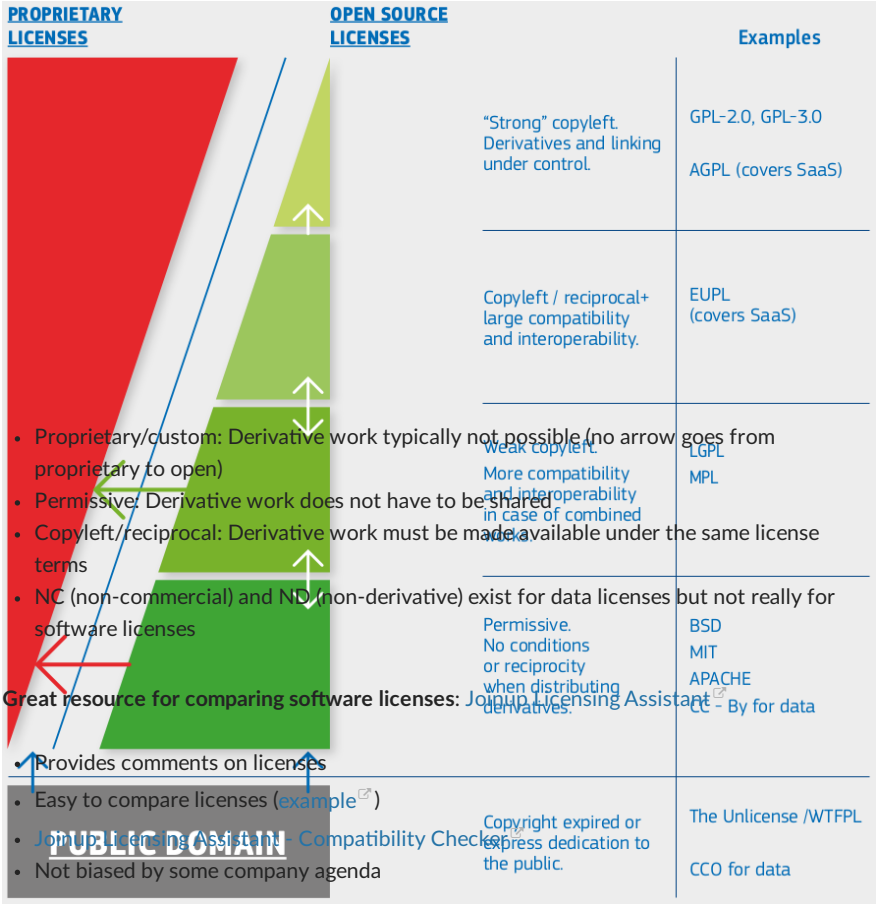
The latter case is a bit more nuanced and the interested reader should read more about "Mere Aggregation" at GPL-FAQ[]. Briefly, if the container image just bundles separate programs that talk through normal system interfaces, it is an **aggregate** and each keeps its own license (like a CD-ROM with various packages). If the components are tightly integrated into one program (e.g. a pipeline with various parts that the container can run as a single program), the image may be treated as a **derivative work**, and stricter license obligations (e.g. GPL copyleft) can apply.

Taxonomy of software licenses



European Commission, Directorate-General for Informatics, Schmitz, P., European Union Public Licence (EUPL): guidelines July 2021, Publications Office, 2021, https://data.europa.eu/doi/10.2799/77160[]

- Proprietary/custom: Derivative work typically not possible (no arrow goes from proprietary to open)
- Permissive: Derivative work does not have to be shared
- Copyleft/reciprocal: Derivative work must be made available under the same license terms
- NC (non-commercial) and ND (non-derivative) exist for data licenses but not really for software licenses

Great resource for comparing software licenses: Joinup Licensing Assistant[]

- Provides comments on licenses
- Easy to compare licenses (example[])
- Joinup Licensing Assistant - Compatibility Checker[]
- Not biased by some company agenda

If you would like to learn more about licenses, check out our slide deck: "Software licensing and open source explained with cakes"[]

Exercise: Licensing situations

Comments:

✍ Licensing-2: Consider some common licensing situations

1. What is the StackOverflow license for code you copy and paste?

In academic context it is important to consider also *plagiarism* and how it relates to copyright and more broadly Intellectual Property Rights (a clear explanation at this page[]). Plagiarism is the practice of taking somebody else's ideas or work and claim them as your own: it is the **unacknowledged** use of another person's work. Intellectual Property Rights (IPRs) infringement instead is the **unauthorised** use of another's work.

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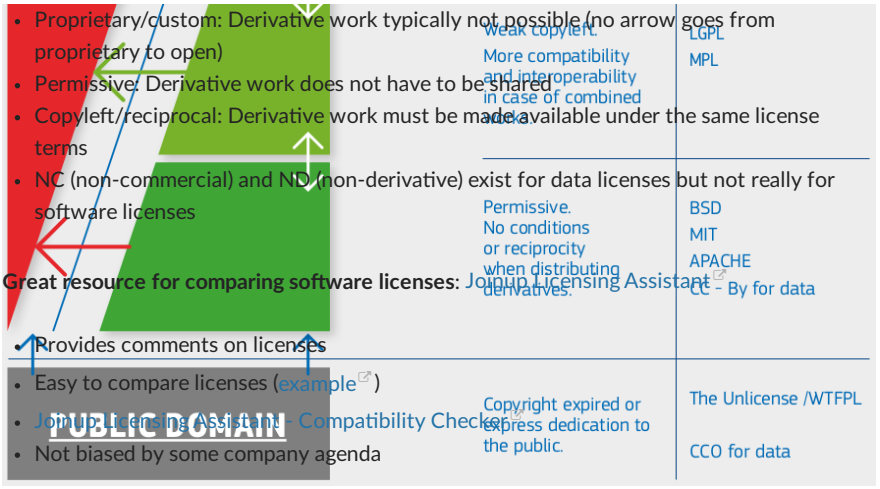

European Commission, Directorate-General for Informatics, Schmitz, P., European Union Public Licence (EUPL): guidelines July 2021, Publications Office, 2021, https://data.europa.eu/doi/10.2799/77160

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Exercise: Licensing situations

Comments:

✍ Licensing-2: Consider some common licensing situations

- Arrows represent compatibility (A -> B: B can reuse A)
- Proprietary/custom: Derivative work typically not possible (no arrow goes from proprietary to open)
- Permissive: Derivative work does not have to be shared
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1. What is the StackOverflow license for code you copy and paste?
2. A journal requests that you release your software during publication. You have copied a portion of the code from another package, which you have forgotten. Can you satisfy the journal's request?
3. You want to fix a bug in a project someone else has released, but there is no license. What risks are there?
4. How would you ask someone to add a license?
5. You incorporate MIT, GPL, and BSD3 licensed code into your project. What possible licenses can you pick for your project?
6. You do the same as above but add in another license that looks strong copyleft. What possible licenses can you use now?
7. Do licenses apply if you don't distribute your code? Why or why not?
8. Which licenses are most/least attractive for companies with proprietary software?

✔ Solution

1. As indicated here, all publicly accessible user contributions are licensed under Creative Commons Attribution-ShareAlike license. See Stackoverflow Terms of service for more detailed information.
2. "Standard" licensing rules apply. So in this case, you would need to remove the portion of code you have copied from another package before being able to release your software.
3. By default you are no authorized to use the content of a repository when there is no license. And derivative work is also not possible by default. Other risks: it may not be clear whether you can use and distribute (publish) the bugfixed code. For the repo owners it may not be clear whether they can use and distributed the bugfixed code. However, the authors may have forgotten to add a license so we suggest you to contact the authors (e.g. make an issue) and ask whether they are willing to add a license.
4. As mentionned in 3., the easiest is to fill an issue and explain the reasons why you would like to use this software (or update it).
5. Combining software with different licenses can be tricky and it is important to understand compatibilities (or lack of compatibilities) of the various licenses. GPL license is the most protective (BSD and MIT are quite permissive) so for the resulting combined software you could use a GPL license. However, re-licensing may not be necessary.
6. Derivative work would need to be shared under this strong copyleft license (e.g. AGPL or GPL), unless the components are only plugins or libraries.
7. If you keep your code for yourself, you may think you do not need a license. However, remember that in most companies/universities, your employer is "owning" your work and when you leave you may not be allowed to "distribute your code to your future self". So the best is always to add a license!
8. The least attractive licenses for companies with proprietary software are licenses where you would need to keep an open license when creating derivative work. For instance GPL and and AGPL. The most attractive licenses are permissive licenses

When should I add a license?

Choose a license early in the project, even before you publish it. Later in the project it may become complicated to change it. Agreeing on a software license does not mean that you have to make it open immediately. You can also follow the **"open core" approach**: You don't have to open source all your work. Core can be open and on a public branch. Unpublished code can be on a private repository.

However, we recommend to **work as if the code is public even though it still may be private** (thanks to E. Glerean for this great suggestion): This is to avoid surprises about code in the history with incompatible license years later when you decide to open the project.

How to add a license if your work is derivative work

Your code is derivative work if you have started from an existing code and made changes to it or if you incorporated an existing code into your code.

If your code is derivative work, then **you need to check the license of the original code**.

Depending on the license, your choices might be limited. In this case we recommend to use

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- Joinup Licensing Assistant - Find and compare software licenses
- Joinup Licensing Assistant - Compatibility Checker

If the original code does not have a license, you may not be able to distribute your derivative code. You can try to contact the authors and ask them to clarify the license of their code.

Practical steps for **incorporating something small into your own project** with a license that allows you to do so (as an example incorporating a function or two from another project):

- Create a `LICENSES/` folder in your project and "put the unmodified license text (i.e., the license text template without any copyright notices) in your `LICENSES/` folder" (https://reuse.software/faq/#license-templates). This way if you reuse code from multiple projects, you can keep there multiple license files.
- **Put the code that you incorporate into a separate file or separate files**. This makes it later easier to see what was incorporated, and what was written from scratch. On top of the file(s) which you have incorporated into your project add (and adapt) the following header (more examples):

```
# SPDX-FileCopyrightText: 2023 Jane Doe <jane@example.com>
#
# SPDX-License-Identifier: MIT
```

- The REUSE initiative was started by the Free Software Foundation Europe to make licensing of software projects easier. It is OK if you prefer to not follow this strict format but the advantage of following it is that the reuse-tool makes it then easy to verify and update license headers if you have many files from different sources.

If your work is not derivative work

If you have started "from scratch", and not used any existing code, or incorporated existing code into your code, then you may consider your code to be not derivative work.

Before you may choose a license, clarify the following points with, for example, your supervisor, collaborators, or principal investigator:

- Does your work contract, grant, or collaboration agreement dictate a specific license?
- Is there an intent to commercialize the code?
- When there is unknown or mixed ownership: If there are multiple persons or organizations as owners of the code, all must agree to the license.

Do not invent your own license. Choose one of the standard licenses, otherwise compatibility is not clear:

- Some licenses are more permissive (you can keep your changes private) but some licenses require you to publish the changes (share-alike).

Practical steps for making **changes to an existing project** with a license that allows you to do so:

- If it does not make sense to have several files in your project (e.g. when incorporating something into a notebook), then add a note/comment about the license and where the code came from on top of the function.
- Although it is not dictated by the license but it can still be nice to acknowledge the incorporated functions/code in your README/documentation and to cite their work if you publish a paper about your code.
- If the project is on GitHub or GitLab or similar, first fork the project (copy it into your user space where you can make changes).

```
# SPDX-License-Identifier: MIT
```

helpful for others if you do. You can state your changes in the header of the files you have modified. It can be helpful to state bigger-picture changes in the README file of the project.

- Some licenses are more permissive (you can keep your changes private) but some licenses require you to publish the changes (share-alike).
- If it does not make sense to have several files in your project (e.g. when incorporating something into a notebook), then add a note/comment about the license and where the code came from on top of the function.
- Although it is not dictated by the license but it can still be nice to acknowledge the incorporated functions/code in your README/documentation and to cite their work if you publish a paper about your code.
- Some licenses are more permissive (you can keep your changes private) but some licenses require you to publish the changes (share-alike).

Practical steps for making **changes to an existing project** with a license that allows you to do so:

- If the project is on GitHub or GitLab or similar, first fork the project (copy it into your user space where you can make changes).

With generative AI tools for coding such as GitHub copilot, Cursor, or even basic chat implementations (ChatGPT, Claude, Grok, …) the responsibility fully lays on the person who is going to use (and publish) the generated code. You can never blame the autopilot or the company who invented it, only the driver (you!).

There are various risks in using generative AI code (this is not a taxonomy). A few examples:

- Risks for the derivative work: you think your code is doing what you asked, but you did not review it and your results are false
- Risks for the system in use: your generated code has software security issues, e.g. an import is a *typosquat* of an actual library (e.g. “microsoft” is spelled “rnicrosoft” and depending on the font you might totally miss it…)
- Risks related to licenses/IPR: you have generated code that is actually verbatim copy of fully copyrighted code, or code that requires a strict copyleft license. Plagiarism (ethics) also applies.

If we focus on the last one, a recent paper (ref) estimates that around 2% of AI generated code is “strikingly similar to existing open-source implementations”. Generative AI tools are typically not able to provide an exact reference of where certain bits of generated code were copied from, so it is the responsibility of the researcher to verify that the produced code is citing and referencing the license of other published pieces of software. Possibly, future AI systems for code generation can be trained on code that share the same set of licenses (e.g. based only on MIT) to mitigate these risks.

The REUSE initiative was started by the Free Software Foundation Europe to make licensing of software projects easier. It is OK if you prefer to not follow this strict format but the advantage of following it is that the reuse-tool makes it then easy to verify and update license headers if you have many files from different sources.

If your work is not derivative work

If you have started “from scratch”, and not used any existing code, or incorporated existing code into your code, then you may consider your code to be not derivative work.

Before you may choose a license, clarify the following points with, for example, your supervisor, collaborators, or principal investigator:

- Does your work contract, grant, or collaboration agreement dictate a specific license?
- Is there an intent to commercialize the code?
- When there is unknown or mixed ownership: If there are multiple persons or organizations as owners of the code, all must agree to the license.

Do not invent your own license. Choose one of the standard licenses, otherwise compatibility is not clear:

- Joinup Licensing Assistant - Find and compare software licenses
- Joinup Licensing Assistant - Compatibility Checker

Practical steps:

- Create a `LICENSES/` folder (example).
- Put the unmodified license text (i.e., the license text template without any copyright notices) in plain text format into the folder (example). Here are the two above licenses in plain text: EUPL and MIT (but the latter contains a copyright notice which we rather want to have on top of files).
- Add copyright and license information to each file following https://reuse.software/tutorial/ which uses a standard format with so-called SPDX identifiers. Example below (example):

```
# SPDX-FileCopyrightText: 2023 Jane Doe <jane@example.com>
#
# SPDX-License-Identifier: EUPL-1.2
```

The REUSE initiative was started by the Free Software Foundation Europe to make licensing of software projects easier. It is OK if you prefer to not follow this strict format but the advantage of following it is that the reuse-tool makes it then easy to verify and update license headers if you have many files from different sources.

- For really small projects with one or two files the above may seem excessive and some projects choose to not have copyright information on top of their files and they only have one `LICENSE` file and that is OK for really small projects.

Great resources

- Research institution policies to support research software (compiled by the Research Software Alliance)
- Guide from the Aalto University in Finland: “Opening your Software at Aalto University”
- Draft: Research software licensing guide
- Joinup Licensing Assistant - Find and compare software licenses
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but the advantage of following it is that the `reuse-tool` makes it then easy to verify and update license headers if you have many files from different sources.

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- Joinup Licensing Assistant - Find and compare software licenses
- Joinup Licensing Assistant - Compatibility Checker
- Social coding lesson material by CodeRefinery
- Citation File Format (CFF)
- License Selector
- GitHub licensing guide
- Choosing an open-source licence
- Understanding Open Source and Free Software Licensing
- Software Licenses in Plain English
- Don's Bibliography of Ethical Source Reading and Resources
- Mikko Välimäki: The Rise of Open Source Licensing
- Lawrence Rosen: Open Source Licensing
- Aalto IPR Cheatsheet
- Contributor License Agreements
- 4OSS recommendations
- 4OSS lesson
- Intellectual Property Rights (IPR), Licensing And Patents
- Dispelling Open Source Confusion: An Introduction to Licenses
- https://choosealicense.com (can send automatic pull request to your GitHub repo)
- https://hintjens.gitbooks.io/social-architecture/content/chapter2.html
- http://rkd.zgib.net/scicomp/open-science/open-science.html
- Nadia Asparouhova (formerly Nadia Eghbal): "Working in Public: The Making and Maintenance of Open Source Software" (Stripe Press)
- Open Source Guides
- The Architecture of Open Source Applications
- Christopher M. Kelty: "Two Bits: The Cultural Significance of Free Software" (Duke University Press, 2008)
- Open Source (Almost) Everything
- 99 ways to ruin an open source project
- Open Source Casebook

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- **You cannot ignore licensing**: default is "no one can make copies or derivative works".

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Authors	CodeRefinery, Radovan Bast, Kirsten Thor Wikfeldt, Johan Hellevik, Richard Darst, Sabry Razick, Jyry Suvilehto, Anne Fouilloux, Oxana Smirnova, Lucy Whalley, Enrico Glerean
Version	2026.08-10
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- Nadia Asparouhova (formerly Nadia Eghbal): "Working in Public: The Making and Maintenance of Open Source Software" (Stripe Press)
- Open Source Guides
- The Architecture of Open Source Applications
- Christopher M. Kelty: "Two Bits: The Cultural Significance of Free Software" (Duke University Press, 2008)
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